# HOMOMORPHISMS INTO A FINITE REFLEXIVE MODULE OVER A NOETHERIAN DOMAIN ARE REFLEXIVE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

*Editorial note.* The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

*Allowed standard prerequisites (editorial).*

- Reflexive module over a Noetherian domain, defined through the double dual map. definition at source lines 5148-5157 of the same chapter. Supplies the hypothesis on the target module.
- A module sitting in a short exact sequence with reflexive middle term and torsion free quotient is reflexive. the author cites Lemma lemma-sequence-reflexive at proof line 5363; that statement is counted separately by this lane. Concludes reflexivity of the module of homomorphisms.
- Dual of a presentation and torsion freeness of submodules of sums of torsion free modules. used at proof lines 5356-5362. Produces the exact sequence and checks that its quotient term is torsion free.

*Source reference.* Original lemma 0AV3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13671–13671. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 0EB8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17913–17913. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $R$ be a Noetherian domain. Let $M$ be a finite $R$-module. Let $N$ be a finite reflexive $R$-module. Then* $\mathrm{Hom}_R(M, N)$ *is reflexive.*

*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* Choose a presentation $R^{\oplus m} \to R^{\oplus n} \to M \to 0$. Then we obtain

$$0 \to \mathrm{Hom}_R(M, N) \to N^{\oplus n} \to N' \to 0$$

with $N' = \mathrm{Im}(N^{\oplus n} \to N^{\oplus m})$ torsion free. We conclude by Lemma 0EB8. $\square$